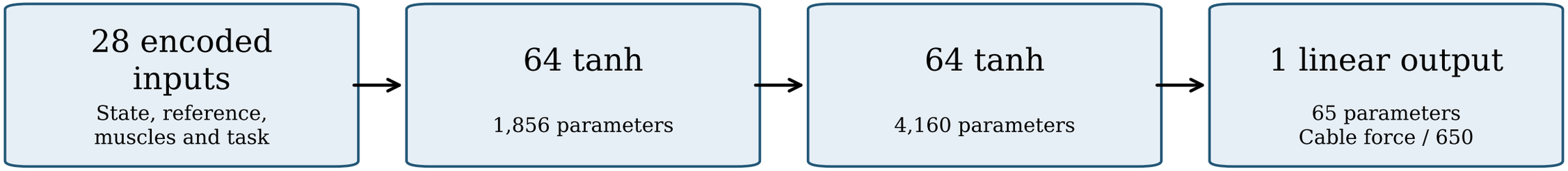
Direct-force neural network: 6,081 trainable parameters



Requested tension is clipped to 0–650 N. The NN supplies learned feedback; the PID teacher labels training data.

$$\ell_w(D) = \frac{\sum_{i \in D} w_i \left[f_\theta(x_i) - F_i/650 \right]^2}{\sum_{i \in D} w_i}$$

$$L_{\text{ordinary}} = \ell_w(D_t) + 0.1R(D_t)$$

$$L_{\text{CL}} = L_{\text{ordinary}} + \beta[\ell_w(\mathcal{M}_A) + 0.1R(\mathcal{M}_A)]$$

$$R(D) = \frac{1}{2K} \sum_{i=1}^{K} \sum_{k \in \{q, v\}} \max(0,\, 0.02 - \widehat{d}_{ik})^2$$

Current-task data: D_t
Old A examples: $\mathcal{M}_A$
Fixed memory: 1,024 examples
Replay: at most512/update
β: 1,4,16 (validation)
$K \le 128$, $\varepsilon = 0.05$